# Weighting Scheme: Inverse Average-Correlation

MetaFVG Diversified Mean-Reversion Portfolio

## 1 Scheme Description

The 14-instrument MetaFVG mean-reversion portfolio (Spearman-gated, `fixed` sizing variant) is deliberately assembled to be fundamentally diversified across central banks and currencies: AUDUSD, EURUSD, GBPUSD, NZDUSD, USDCAD, USDCHF, USDJPY, US500, GER40, JP225, HK50, USDSGD, EURNOK, USDZAR. It is currently blended with equal weights (1/14 each). This report tests **Inverse Average-Correlation** weighting (`inverse_avg_corr`): the most literal reading of "target uncorrelated results" — weight each instrument inversely proportional to how correlated it is, on average, with the rest of the book.

For each instrument $s$, let $\rho_{s,x}$ denote the pairwise Pearson correlation of daily P&L returns between $s$ and instrument $x$ (computed pairwise-complete over the full return history). The average correlation of $s$ with the rest of the book is

$$\overline{\rho}_s = \frac{1}{n-1} \sum_{x \neq s} \rho_{s,x}, \qquad n = 14.$$

Because the portfolio was already fundamentally diversified by design, these average correlations are small and can be near zero or slightly negative — naively inverting such a value blows up. A floor is applied before inversion:

$$d_s = \max(\overline{\rho}_s,\ 0.02).$$

Raw weights are then

$$\tilde{w}_s \propto \frac{1}{d_s}, \qquad w_s^{(0)} = \frac{\tilde{w}_s}{\sum_x \tilde{w}_x}.$$

Finally, a concentration cap is applied at $3\times$ the equal weight ($3/14 \approx 0.2143$): any weight exceeding the cap is clipped to it, and the remaining (uncapped) weights are renormalized proportionally so that all weights still sum to 1.

**Safeguards triggered:** the floor bound for **9 of 14** instruments — EURNOK, EURUSD, HK50, JP225, USDCAD, USDCHF, USDJPY, USDSGD, USDZAR — all of which had raw average correlation below 0.02 (three of them, EURNOK, USDCHF and HK50, were in fact very close to zero or negative). These instruments were all floored to the same denominator (0.02) and therefore received an identical final weight ($\approx 0.0811$, or $1.136\times$ equal weight). The 3x concentration cap (0.2143) did **not** bind for any instrument – the highest resulting weight was only $\approx 1.14\times$ equal weight, well under the cap.

Table 1: Per-instrument average correlation, resulting weight, and weight relative to equal weight ($1/14 \approx 0.0714$), sorted descending by weight. F = floor bound, C = cap bound.

| Symbol | Avg. Corr. ($\overline{\rho}_s$) | Weight | Weight / (1/14) | Safeguard |
|---|---|---|---|---|
| EURUSD | 0.0196 | 0.0811 | 1.14× | F |
| USDCAD | 0.0168 | 0.0811 | 1.14× | F |
| USDCHF | 0.0004 | 0.0811 | 1.14× | F |
| USDJPY | 0.0056 | 0.0811 | 1.14× | F |
| JP225 | 0.0188 | 0.0811 | 1.14× | F |
| HK50 | 0.0016 | 0.0811 | 1.14× | F |
| USDSGD | 0.0120 | 0.0811 | 1.14× | F |
| EURNOK | −0.0153 | 0.0811 | 1.14× | F |
| USDZAR | 0.0070 | 0.0811 | 1.14× | F |
| NZDUSD | 0.0265 | 0.0613 | 0.86× | – |
| US500 | 0.0276 | 0.0588 | 0.82× | – |
| AUDUSD | 0.0307 | 0.0528 | 0.74× | – |
| GER40 | 0.0313 | 0.0519 | 0.73× | – |
| GBPUSD | 0.0360 | 0.0450 | 0.63× | – |

## 2 Weights

## 3 Correlation Impact

Weighted average pairwise correlation is $\left(\sum_{i\neq j} w_i w_j \rho_{ij}\right) / \left(\sum_{i\neq j} w_i w_j\right)$; for equal weights this reduces to the simple average of the off-diagonal correlation matrix ($\approx 0.0156$). Diversification ratio is $\mathrm{DR} = \left(\sum_i w_i \sigma_i\right) / \sigma_{\text{portfolio}}$, using each instrument's own-sample standard deviation of daily returns and the standard deviation of the final blended daily-return series.

Table 2: Correlation impact: equal weight vs. inverse-average-correlation weight.

| Metric | Equal Weight | Inverse Avg. Corr. |
|---|---|---|
| Weighted avg. pairwise correlation | 0.0156 | 0.0118 |
| Diversification ratio (DR) | 2.081 | 2.110 |

## 4 Results

Both blends use per-day NaN-aware renormalization: on days where some instruments have no closed trade (NaN return), the scheme's weights are renormalized to sum to 1 over only the instruments with a non-NaN return that day. Days with all-NaN returns are dropped. Both series contain 1,661 daily observations after this filtering. Quantstats metrics are computed on the resulting equity curve (daily resampled).

Table 3: Performance metrics: equal weight vs. inverse-average-correlation weight.

| Metric | Equal Weight | Inverse Avg. Corr. |
|---|---|---|
| Sharpe | 0.518 | 0.483 |
| Sortino | 0.817 | 0.760 |
| Calmar | 0.343 | 0.315 |
| CAGR | 0.0183% | 0.0168% |
| Max Drawdown | −5.33% | −5.33% |
| Volatility (ann., %) | 3.528% | 3.473% |
| Tail Ratio | 1.239 | 1.288 |
| Ulcer Index | 0.000192 | 0.000173 |
| Kelly | 0.0668 | 0.0610 |
| VaR (%) | −0.358% | −0.353% |
| CVaR (%) | −0.715% | −0.692% |
| Skew | 1.545 | 1.544 |
| Recovery Factor | 2.260 | 2.075 |
| Win Rate | 30.89% | 30.52% |
| Best Day | 0.0291% | 0.0291% |
| Worst Day | −0.0239% | −0.0239% |
| Total Return | 0.1204% | 0.1105% |

## 5 Takeaway

Inverse average-correlation weighting achieved the intended correlation effect: weighted average pairwise correlation fell from 0.0156 (equal weight) to 0.0118, and the diversification ratio improved modestly from 2.081 to 2.110. Neither safeguard was actually a problem in practice — the floor caught 9 of 14 instruments (clustering them at an identical weight), while the 3x cap never bound, meaning the floor did all the real work of shaping this allocation. However, the reduction in realized correlation did *not* translate into better risk-adjusted returns: Sharpe fell from 0.518 to 0.483 and Sortino, Calmar, Kelly, and Recovery Factor all declined in tandem. The reason is visible in the individual-instrument statistics: the scheme's three lowest final weights went to GBPUSD, AUDUSD and NZDUSD (annualized single-instrument Sharpe 0.76, 0.66 and 0.76 respectively — three of the book's best performers), while the floor-driven cluster of nine equally-weighted instruments includes EURUSD, USDCAD, USDSGD, EURNOK and USDZAR, all of which have *negative* single-instrument Sharpe on their own. Low average correlation with the rest of the book and being a strong individual performer are simply different properties, and this scheme optimizes only for the former. On this dataset it trades away Sharpe for a small diversification-ratio gain — a bad trade for a book that was already reasonably diversified by construction.